Conditional-Capture Flip-Flop for Statistical Power Reduction

Bai-Sun Kong, Sam-Soo Kim, and Young-Hyun Jun

***Abstract*—This paper describes a family of novel low-power flip-flops, collectively called conditional-capture flip-flops (CCFFs). They achieve statistical power reduction by eliminating redundant transitions of internal nodes. These flip-flops also have negative setup time and thus provide small data-to-output latency and attribute of soft-clock edge for overcoming clock skew-related cycle time loss. The simulation comparison indicates that the proposed differential flip-flop achieves power savings of up to 61% with no impact on latency while the single-ended structure provides the maximum power savings of around 67%, as compared to conventional flip-flops. With a typical switching activity of 0.33, the power consumption is reduced by as much as 23–30% with comparable minimum data-to-output latency. It is also indicated that the proposed single-ended structure provides power comparable to the fully static master–slave design with significantly reduced data-to-output latency. An eight-bit counter was fabricated using a 0.35-μm CMOS technology, and the experimental results indicate that the counter using the differential CCFF saves the overall power consumption by about 30% as compared to that using the conventional flip-flop.**

***Index Terms*—CMOS digital integrated circuits, flip-flops, low-power, master-slave latches.**

I. INTRODUCTION

SYNCHRONOUS digital systems incorporate the clock signal(s) for controlling the execution of their sequential parts. A number of different clocking strategies can be used for implementing such systems, along with several different types of storage elements such as flip-flops and latches. The clocking schemes for these systems must be as robust and easy to implement as possible. The single-phase and two-phase clocking schemes are the most commonly encountered. A detailed analysis of the timing requirements of various clocking schemes for synchronous digital systems is discussed in [1]. The paper develops sets of relations for making tradeoffs between the possibility of maximizing the clock frequency and the difficulty of satisfying constraints on the logic path delay. Storage elements such as flip-flops and latches must also be as simple as possible and satisfy various requirements of synchronous systems such as rigid timing constraints and low power dissipation. Timing parameters such as data-to-output latency and setup and hold times are important because the timing budget is getting tighter as the clock frequency increases. Power requirement is also becoming equally important, because the power consumption increases linearly with clock frequency while the power budget of high-performance portable digital systems is severely limited. Some papers provide comparative analysis of latches and flip-flops commonly used in high-performance systems and deal with the possibilities of trading off one for the other between speed and power [2], [3]. In addition to these utmost important parameters, there are a variety of other parameters related to latching elements, which can have a certain impact on the performance of synchronous systems. The ability to absorb the clock skew is attaining wide attention to minimize the impact of the clock skew that occupies a nonnegligible fraction of a cycle time as the clock frequency increases. The clock load imposed by flip-flops and latches is important because in high-performance digital systems the power consumption contributed by the clock can be as much as 45% of the overall power consumption [4]. The integration of logic into the latching element is also important as the depth of the logic level per pipeline stage is reduced due to the shrinking of cycle time.

Recently reported flip-flops such as hybrid latch-flip-flop (HLFF) [5], semi-dynamic flip-flop (SDFF) [6], and sense amplifier-based flip-flop (SAFF) [7] [8] are commonly used for implementing high-performance digital systems due to small delay between the latest point of data arrival and the corresponding output transition. They can also provide advantages such as absorbing the clock skew, reducing the clock load, and embedding logic functions into themselves. But, as will become clear in the next section, they are inefficient as far as power consumption is concerned. This is attributed to the operation principle that, in some situations, these flip-flops can have unnecessary internal transitions that can lead to a substantial increase in overall power dissipation. This paper describes a family of newly developed flip-flops that can eliminate these redundant transitions to minimize power dissipation with no impact on latency [9]. In Section II, we present the analysis of conventional high-performance flip-flops and the discussion of the drawbacks of them in terms of power consumption. Section III introduces the circuit structure and operation principle of the proposed flip-flops. Section IV presents the performance comparison of several flip-flops, and explains the design of an eight-bit counter to highlight the low power advantage of the proposed flip-flops. Finally, we draw our conclusion in Section V.

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B.-S. Kong is with the School of Electronics Telecommunications and Computer Engineering, Hankuk Aviation University, Goyang, Kyunggi-do, 412-791 Korea (e-mail: baiskong@hotmail.com).

S.-S. Kim is with Hyundai Electronics (currently Hynix Semiconductor), Seoul, 135-738, Korea (e-mail: kiss@hei.co.kr).

Y.-H. Jun is with the School of Electrical and Computer Engineering, Sungkyunkwan University, Suwon, Kyunggi-do, 440-746, Korea (e-mail: yhjun@skku.ac.kr).

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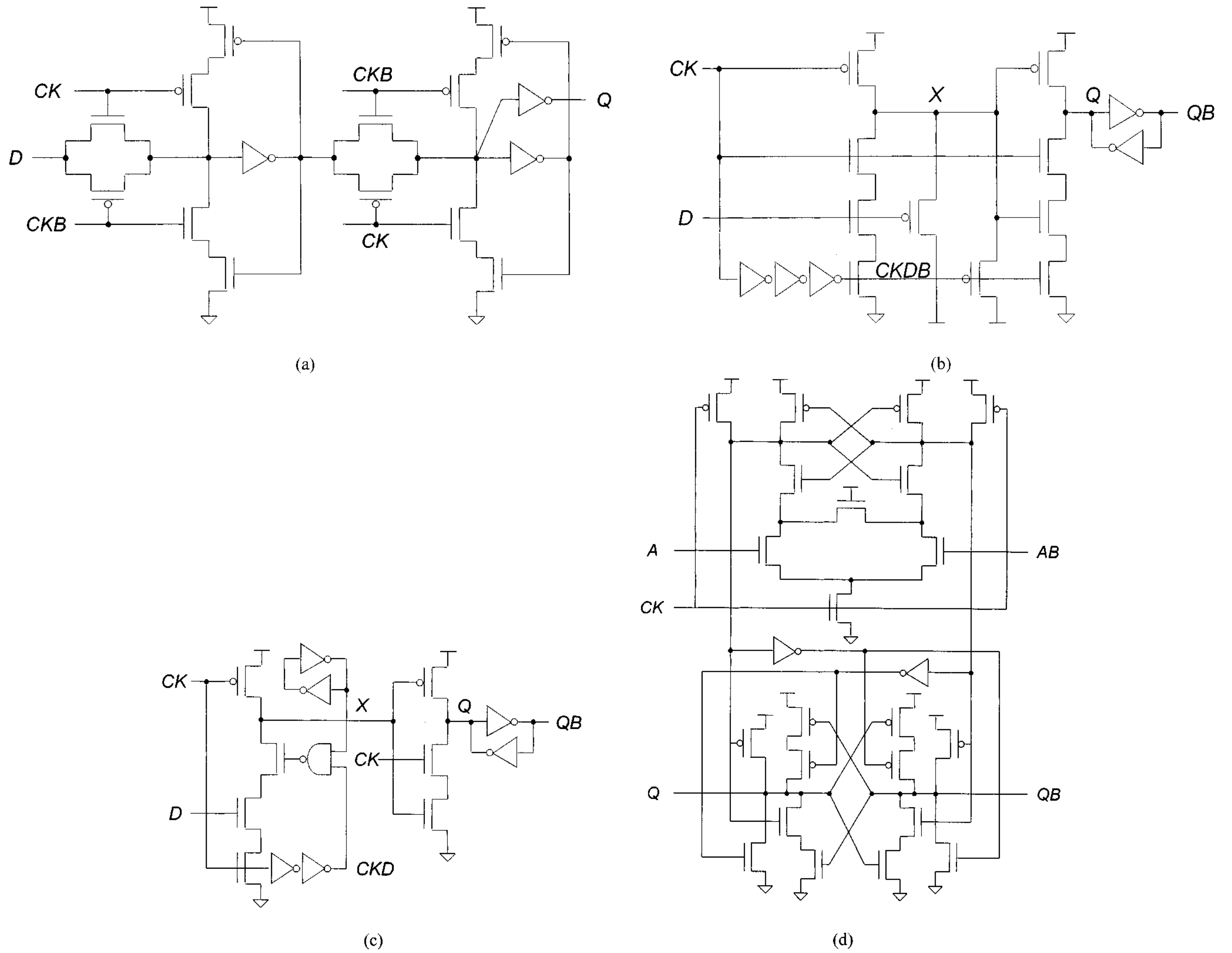

Fig. 1. Conventional high-performance flip-flops: (a) TGFF, (b) HLFF, (c) SDFF, and (d) SAFF.

II. Power Consumption of Flip-Flops

Fig. 1 shows the schematic diagram of representative high-performance flip-flops. They are transmission-gate flip-flops (TGFFs) [10], hybrid latch-flip-flops (HLFFs), semi-dynamic flip-flop (SDFF)s, and sense amplifier-based flip-flops (SAFFs). TGFFs have a fully static master–slave structure by cascading two identical pass-gate latches and provides a short clock-to-output latency. However, it has poor data-to-output latency because of positive setup time. It also requires two phases of clock that can cause a problem with data feedthrough when there is a skew between them and it has a relatively large clock load. HLFFs and SDFFs are actually latches with a brief transparency period. They implement the idea of a hybrid design technique in which the transparency period of a latch is reduced to a small interval. The major advantage of the hybrid design is the soft-clock edge property, i.e., robustness to clock skew, while the major drawback is that the hold time has a positive value, mandating a detailed hold timing analysis to avoid a timing failure. SDFF employs a conditional shutoff capability to provide a reduced sensitivity to the variations of the sampling window, or to provide narrower sampling window for shorter hold time. SAFF incorporates the precharged sense amplifier in the first stage to generate a negative pulse, and a set–reset (SR) latch in the second stage to capture the pulse and hold the results. The latency of NAND-based SAFFs [7] is longer than that of HLFFs because one output is delayed by one gate delay from the other, which was overcome by a recently published SAFF having the symmetric slave latch [8]. They are faster than the standard master–slave structure and provide hard edges. The latter also allows fully symmetrical output transitions. These differential flip-flops and forgoing single-ended structures such as the HLFF and SDFF are called dynamic design as they are operated by the principle of precharging, while the TGFF is called static design as it is implemented using static latches. HLFFs and SDFFs are also referred to as the hybrid design for the reason described above.

Before comparing these flip-flops in terms of power consumption, let us review the sources of the power consumed by a latching element. There are two main sources of power dissipation. One is the internal power dissipation, which represents the portion of the power dissipated by the internal and input nodes during latching operations. This component excludes the power dissipated by the outputs for switching the output loads. The other is the clock power dissipation that represents the portion of the power dissipated by the clock input due to the toggling of the clock signal. The total power consumption refers to the sum of these two components. The clock power dissipation is solely determined by the clock load of a flip-flop or a latch, whereas the distribution of the internal power dissipation is affected by the structure and operation of the latching element itself as well as the input switching activity that is defined as the average number of input transitions per clock cycle.

As for the static design such as a TGFF, the internal power distribution is strongly affected by the statistics of applied input as the transitions of internal nodes are invoked by input data transitions during the transparency period. This implies that with a low input switching activity the internal power dissipation can be very small, whereas the clock power is always large due to the large clock load. Thus, for such a case and similar other cases, the internal power is a negligible fraction of the overall power, making the clock power dominant. As an attempt to reduce this relatively large clock power dissipation, a data-transition look-ahead DFF was proposed to shut down the clock transitions when there is no data transition [11]. HLFF and SDFF identically have the semi-dynamic structure where the first stage is dynamic and the second stage is static. The semi-dynamic nature of these flip-flops provides different internal power dissipation dependencies on input data distribution as compared to a fully static master–slave structure. This is because the precharge node of the dynamic stage is charged and discharged during each clock cycle even if the input data remains at a high logic value. Thus, hybrid designs can have sustained internal power dissipation for some input patterns due to the precharge nature of the front-end stage. Because these internal transitions invoke no output transitions during this period, this portion of the power is apparently redundant. The clock power dissipation is smaller than the static design because it uses a single-phase clock and the clock load is relatively small. SAFF has the differential dynamic structure producing a monotonous pull-down transition at the rising edge of the clock. Since this transition always occurs at one side of the differential stage at each clock cycle regardless of input data pattern, this type of flip-flop provides relatively data-independent internal power distribution. As in an HLFF, a large portion of the power can be redundant if the transition does not result in an output transition. It has relatively small clock power as in the hybrid design.

In summary, the hybrid and differential designs are faster but consume more power than the static counterpart especially for low input switching activities. Thus, for systems where high performance is of prime concern within a given power budget, hybrid designs can be considered to be a good solution, given their negative setup times and resulting small latencies. Static designs are suitable for low-power applications, where speed is of secondary concern. But, if both the speed and power are equally important, it seems that we have no option to satisfy these two conflicting requirements simultaneously. However, it may be possible if there is a hypothetical circuit that can provide a timing performance comparable to the hybrid design with power dissipation comparable to the static design. One fortunate fact here is that a fairly large fraction of internal power consumption of hybrid and differential designs can be redundant for particular input patterns as explained earlier and, thus, can be eliminated if there is a proper measure. We focus on this aspect and introduce a family of novel flip-flops that can achieve this goal in the next section.

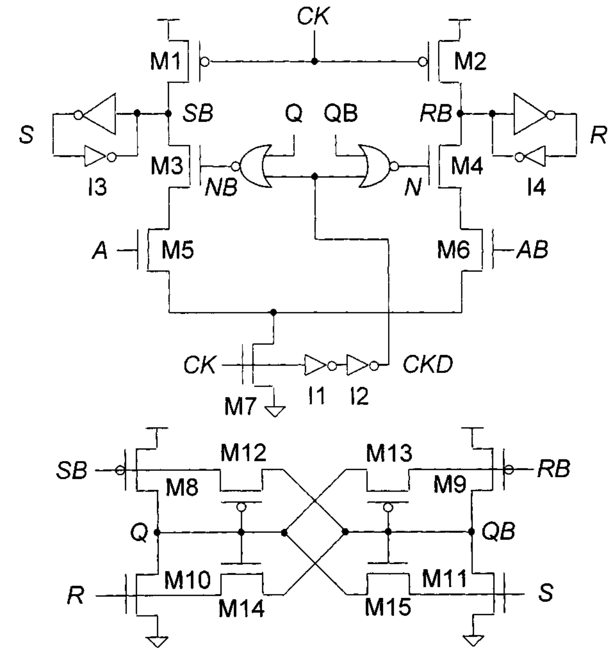

Fig. 2. Conditional capture flip-flop (differential).

III. Conditional-Capture Flip-Flops

The schematic diagram of the proposed flip-flop, the conditional-capture flip-flop (CCFF), is shown in Fig. 2. The flip-flop consists of two stages, the differential circuit with a pair of NOR gates and clock inverters I1 and I2 in the first stage, and the high-speed SR latch with the cross-coupled circuit in the second stage. The NOR gates are driven by the outputs to make the discharge of precharge nodes, SB and RB, depend on the equivalence of input and output data. They are also driven by the delayed version of the clock, CKD, to determine the transparency period. The outputs, N and NB, of the NOR gates drive the transistors, M4 and M3, of the pull-down paths, respectively. Two weak inverters, I3 and I4, are used to maximize the noise immunity of precharge nodes by making them fully static. Two pairs of outputs, S/SB and R/RB, from the first stage are fed into the second stage, where the SR latch is stationed. The latch captures each transition and holds the outputs until the next pull-down transition occurs on one of the precharge nodes. The cross-coupled circuit consisting of four weak transistors M12 through M15 is used for compensating the leakage and preserving the output data statically during the period in which the SR latch is opaque.

The operation of the flip-flop is as follows. When the clock is low, SB and RB are precharged to the supply voltage through the precharge transistors M1 and M2. The internal nodes S and R are pulled down to the ground. With these four nodes fed into the second stage, all the transistors M8 through M11 in the SR latch become fully off, which implies that the latch is in the opaque phase. If it is assumed that the flip-flop outputs Q and QB were evaluated to have low and high logic values, respectively, during the preceding clock cycle, the transistors

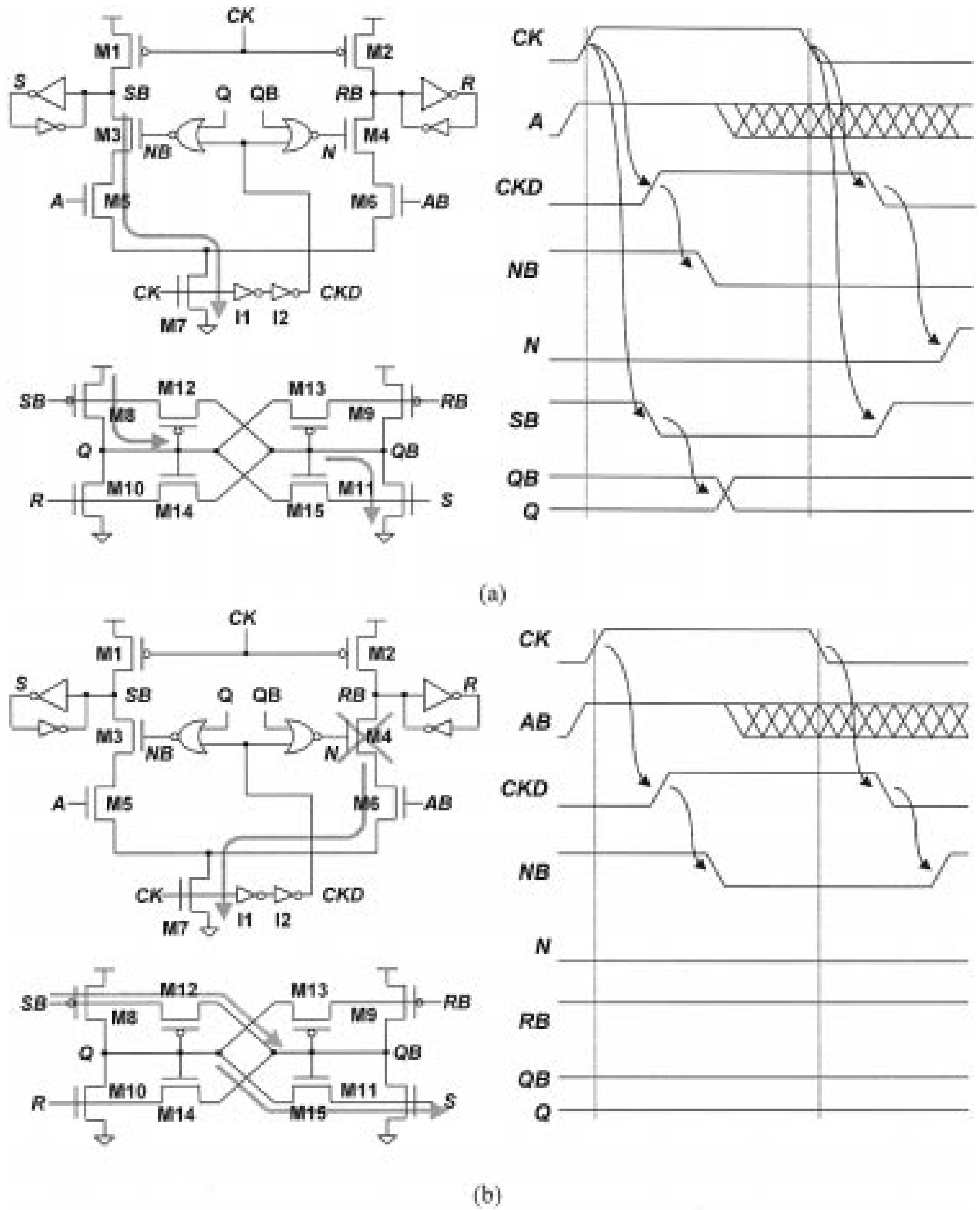

Fig. 3. Operation of the differential CCFF: (a) for a different input value from that of the output and (b) for the same input value as that of the output.

M12 and M15 of the cross-coupled circuit become active to preserve the output logic states while the transistors M13 and M14 are in the cut-off region. Meanwhile, the outputs of NOR gates N and NB are at low and high logic values, respectively, turning M3 on and M4 off. (Notice that CKD is also low because the input clock CK is still at a low logic value.) Because these transistors are in different operating regions, the operation of the flip-flop after the rising clock edge is determined by the incoming data. Let us first explain the case in which the applied input value is different from that of the output. The circuit operation for this case is represented in Fig. 3(a). With a high logic value applied to the input A (recall that the output Q was assumed to be low), SB is pulled down at the rising clock edge because M3 and M5 are simultaneously on. This, in turn, causes the internal node S to be pulled up. Then, the transistors M8 and M11 become active to pull up the output Q and pull down the output QB, respectively. During the period in which these transitions occur, the transistors M12 and M15, which were active before the rising clock edge, become off, causing no signal fighting between the old and new values on the outputs. Once CKD transitions high after the delay of the clock inverters, the node NB transitions low and the transistor M3 is turned off, allowing no additional internal node transitions due to subsequent input change. When the clock changes to a low logic value, the precharge transistor M1 again precharges the node SB to the supply voltage. After that, the pull-down of CKD allows the node N instead of NB to be pulled up because the output states have been changed. Then, the transistor M4 turns on, and RB is allowed to be pulled down when an adequate input arrives. SB cannot be pulled down because the transistor M3 stays off. Next, let us consider when the applied input A has a low logic value, equal to that of the flip-flop output Q. In this case, thepull-down

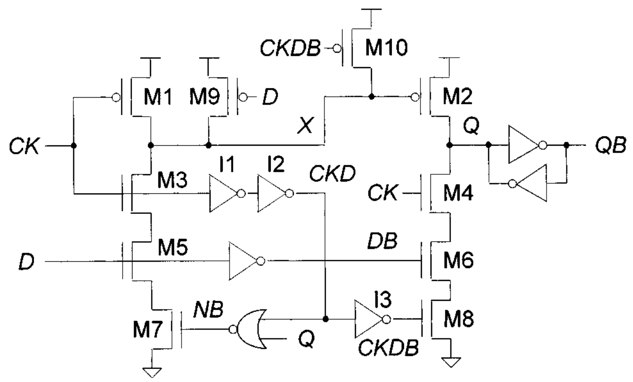

Fig. 4. Conditional capture flip-flop (single-ended).

transition of RB cannot be produced at the rising clock edge because the output of the NOR gate, NB, has a low logic value, turning the transistor M4 off, as shown by Fig. 3(b). Hence, no internal node transitions occur except for the clock path, leading to no output transition. This operation causes no harm because the output Q is at a low logic value, already the same as the one to be latched during the current cycle. As soon as CKD transitions high, the node NB is pulled down to decouple the precharge node from the output, as in the above case. Subsequent pull-down of CKD invoked by the falling edge of the clock makes the node NB pulled up again (note that the outputs has not been changed), and allows the precharge node SB to be discharged during the subsequent cycles.

Fig. 4 shows the single-ended version of the proposed flip-flop. The flip-flop uses the clock inverters and the NOR gate for the same purpose as the differential structure. Unlike the conventional single-ended flip-flops such as HLFF and SDFF described in the previous section, the precharge node X drives only the output pull-up transistor M2 in the second stage. The pull-down transistor M6 is driven by the internal signal DB generated from the input D using an inverter. During the period in which the clock CK is at a low logic value, the precharge node X is precharged to the supply voltage through the pMOS transistor M1. Because CKD is also at a low logic value, the output of the NOR gate NB is conditional to the output Q. That is, if the output is at a low logic value, NB becomes high, and if the output is high, it becomes low. Thus, the precharge node X discharges at the rising clock edge only when Q is at a low logic value and D is at a high logic value, which implies the pull-up transition of the output. For the remaining cases in which Q is at a high logic value or D is at a low logic value, X is held at the precharged voltage because at least M5 or M7 remains off. It is noted that the conventional single-ended flip-flops described earlier have the precharge node pulled down for some of these cases, unnecessarily consuming the switching power. When D is at a low logic value, M6 turns on, and Q stays at or is pulled down to the ground. Once NB transitions low by the pull-up of the delayed version of the clock CKD, the node X is decoupled from the input as in the differential structure. Q is also decoupled from DB when $CKDB$ becomes low. At the falling edge of the clock, X is precharged to or held at the supply voltage and stay there as long as the clock remains low.

The simulated waveforms of the proposed flip-flops are shown in Fig. 5, which are obtained in a 2.5-V 0.35-μm CMOS technology at room temperature with typical devices using 400-fF output loads. These waveforms demonstrate the conditional transition operations of the proposed flip-flops. As for the differential structure, at the first rising edge of the clock the precharge node SB is pulled down to change the output values. However, there are no internal node transitions at the second rising clock edge because the input has not been updated. The single-ended structure also has no transitions on the precharge node at the second rising edge of the clock. The waveform of the differential flip-flop also shows fully symmetrical pull-up and pull-down output transitions.

IV. Comparison and Discussion

As mentioned earlier, the conventional hybrid and differential flip-flops suffer from power consumption due to redundant internal transitions. As for the SAFF, these transitions occur when input is at a high or a low logic value for more than one cycle. The same condition occurs in HLFF and SDFF with input staying at a high logic value. During these periods, conventional flip-flops steadily consume the internal power that they can do without otherwise, increasing the overall power dissipation. This power becomes a major portion when the input has low switching activities. On the contrary, internal precharge nodes of CCFF have transitions only when the outputs need to be changed, as can be recognized by the operation principle described earlier. If the incoming value is identical to the output, the associated precharge node is held high, allowing no internal power dissipation except for the clock path. This leads to a significant power reduction during latching operations, especially for low input switching activities.

To get a more realistic picture, the overall power consumption of several flip-flops for different input patterns is simulated and plotted in Fig. 6. We have applied four different data sequences to represent various input switching activities. The sequence of …01 010 101… represents the maximum input switching activity that reflects the maximum internal power consumption of flip-flops. The switching activity of 0.5 is represented by the sequence of …00 110 011… Two other sequences, …11 111 111… and …00 000 000…, are used to represent the switching activity of zero because each flip-flop has somewhat different power distributions for each of these sequences. For the sake of a fair comparison, the simulation results are separated into two groups, as differential and single-ended. As for the differential, the figure shows that SAFF has relatively a flat power distribution regardless of difference on input patterns while CCFF has a large dependency. For example, CCFF can save 61% power with zero input switching activity as compared to SAFF. When input changes at every other cycle, the power savings is nearly 20%. When the input changes at every clock cycle or the input switching activity is the maximum, which is very rare, the conventional flip-flop is slightly better. The power distribution of the conventional single-ended flip-flops is also flat except for the case in which the input stays at a zero logic value, where they consume inherently small internal power due to no discharge of the precharge node. Single-ended CCFF achieves the maximum power savings of around 67%

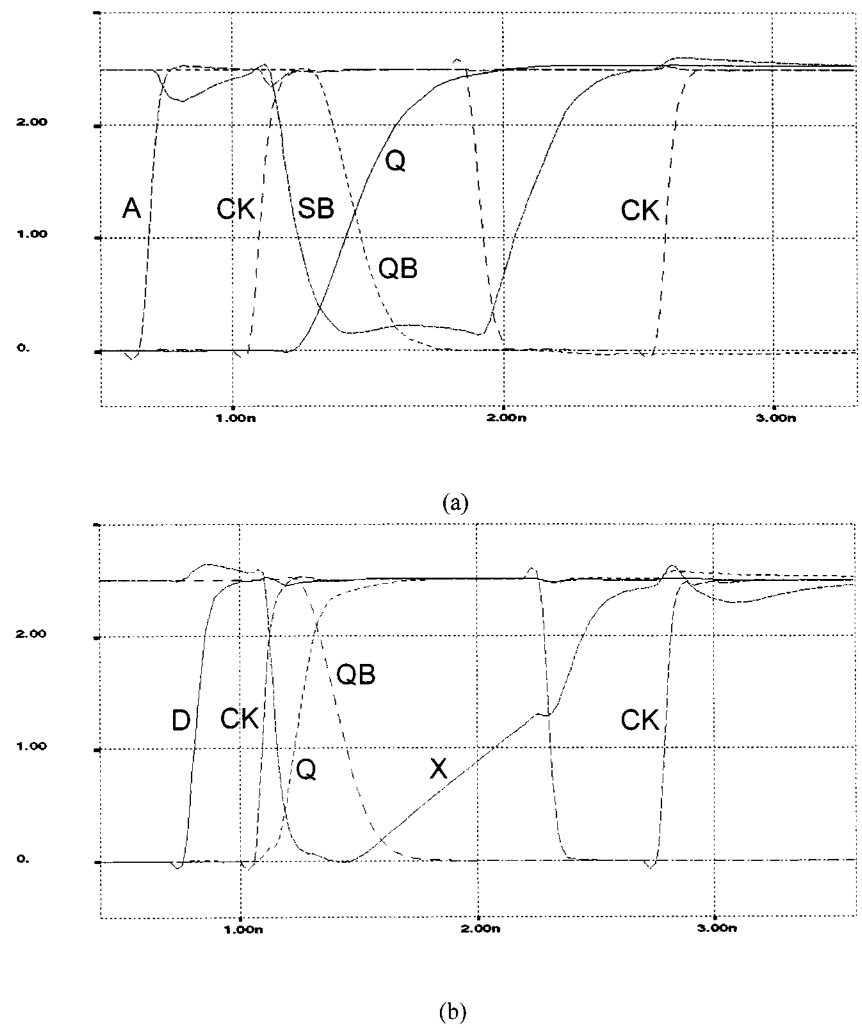

Fig. 5. Simulated waveforms (with the supply voltage: 2.5 V; temperature: 25 °C; capacitive load: 400 fF): (a) differential CCFF and (b) single-ended CCFF.

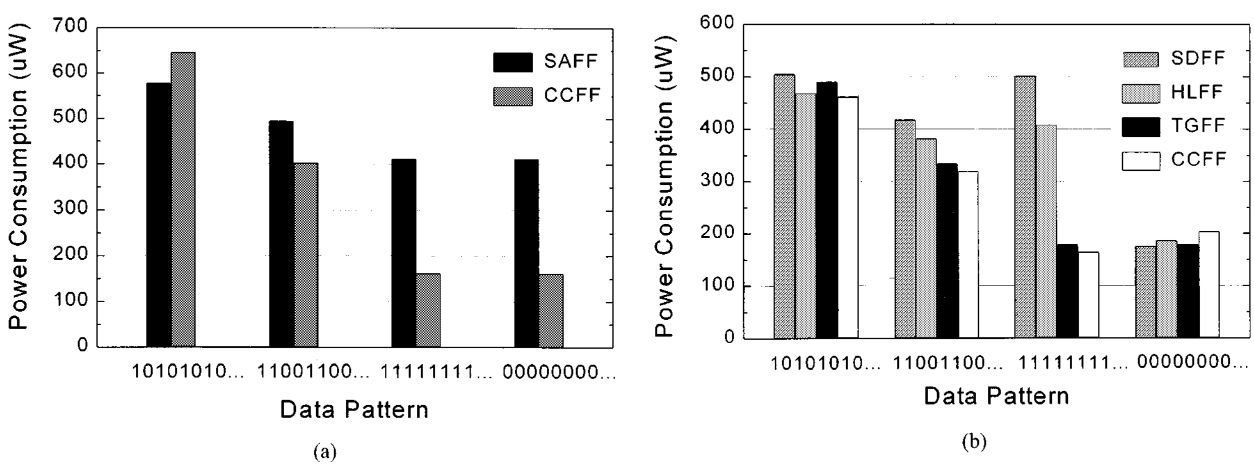

Fig. 6. Power consumption in terms of switching activity (simulated with the supply voltage: 2.5 V, temperature: (a) differential structure, (b) single-ended structure.

with all high inputs as compared to SDFF having the maximum power consumption among those compared. For the maximum switching activity, the overall power consumption is comparable to the conventional designs. It is noted that for all the input patterns the power consumption of the single-ended CCFF is comparable to the TGFF which known to have the least power [2], which implies that the proposed flip-flop is among the flip-flops with the lowest power consumption.

Table I summarizes some important characteristics of flip-flops such as the device count, the clock load, the average power consumption, the minimum data-to-output (D-to-Q) latency, and the corresponding power-delay product. The

TABLE I
FLIP-FLOP CHARACTERISTICS (SIMULATED WITH SUPPLY VOLTAGE: 2.5 V; TEMPERATURE: 25 °C; CAPACITIVE LOAD: 400 fF, CLOCK FREQUENCY: (a) DIFFERENTIAL STRUCTURE AND (b) SINGLE-ENDED STRUCTURE

	Device Count	Total Gate Width(μm)	Clock Load (fF)	Minimum D-to-Q(ps)	Power (μW)	PDP (fJ)	Ratio
SAFF	26	138	30	332	467	155	1
CCFF	35	146	30	337	322	109	0.70

(a)

	Device Count	Total Gate Width(μm)	Clock Load (fF)	Minimum D-to-Q(ps)	Power (μW)	PDP (fJ)	Ratio
HLFF	20	107	27	365	351	128	1
SDFF	23	114	33	361	371	134	1.05
TGFF	18	89	70	433	283	123	0.96
CCFF	26	117	27	356	272	97	0.76

(b)

TABLE II
SETUP AND HOLD TIME CHARACTERISTICS (SIMULATED WITH SUPPLY VOLTAGE: 2.5 V; TEMPERATURE: 25 °C; CAPACITIVE LOAD: 400 fF; CLOCK FREQUENCY: 200 MHz)

	Differential		Single-ended	
	SAFF	CCFF	HLFF	CCFF
Setup time (ps)	-30	-50	-55	-10
Hold time (ps)	180	200	250	230

device count of the proposed flip-flops exceeds that of the conventional ones. But, they have minimal impact on the total gate width as compared to the conventional high-performance flip-flops such as SAFF, HLFF, and SDFF, because all additional devices have a minimum transistor width. For example, the NOR gates can have minimum widths as they are not on the critical path. The device width of the cross-coupled circuit in the second stage of the differential structure is also a minimum as it is used only for compensating the leakage during the opaque phase. It should be noted that the total gate width of the single-ended CCFF is much larger than that of the TGFF. But, in terms of other important performance aspects such as latency and clock load, the former is far better than the latter, as will be shown below. As far as the average power consumption is concerned, which is measured with the average input switching activity of 0.33, the differential CCFF achieves 30% reduction when it is compared to the SAFF. The single-ended version also consumes 23–27% less power than HLFF and SDFF with comparable minimum data-to-output latency. The TGFF has power comparable to CCFF, but the minimum data-to-output latency is far greater due to a positive setup time, leading to poor speed performance. Larger clock load and the need for an inverted clock signal are other important drawbacks of the TGFF. The proposed flip-flops can also achieve further power reduction by using reduced clock swing operation [12]. They also can integrate logic function into the flip-flop and provide fully symmetrical output transitions for the differential structure, as the conventional ones do.

The setup and hold time characteristics of the flip-flops are summarized in Table II. The proposed flip-flops have negative setup time and, thus, can provide important attributes of soft-clock edge for borrowing cycle time to help balance the timing budget between adjacent pipeline stages and for overcoming a loss of useful cycle time due to clock skew. For the single-ended structure, the setup time cannot be deeply negative because the voltage dip on Q may not be immediately recovered. The negative setup time attribute of the CCFF is provided by a brief transparency period, which is defined by the timing period between the rising clock edge and the falling edge of the NOR output. Thus, the width of the transparency period is determined by the delays of the inverters and the NOR gate. If this width becomes too narrow due to any possible variations on the process and operating conditions, the precharge node that must go low cannot be fully discharged to the ground level and, thus, the flip-flop may not correctly evaluate the input or may cause static power dissipation. Hence, the width of the transparency period must be carefully determined with proper timing margin and guaranteed for all the possible worst-case process and operating conditions. The hold time of the proposed flip-flops is also affected by this period. Because it usually has a positive value as shown by Table II, the design incorporating the CCFF requires a detailed hold timing analysis as in the conventional hybrid designs.

Because the internal power consumption of the proposed flip-flops has strong dependency on the switching activity of input data as shown above, their advantage of low power can be best exploited by the applications in which data switching activity is inherently low. For applications in which internal nodes are inherently very busy, and thus, have transitions at almost every cycle, the proposed design may not provide a desired result. This is because the power consumption of the CCFF with maximum input switching activity is comparable to that of conventional ones for the single-ended design, and even higher for the differential design. But, because the switching activity of random data is 0.5 and a great deal of ordinary applications have data switching activity well below this value [13], the proposed design can provide a substantial power reduction for many ordinary applications. To achieve

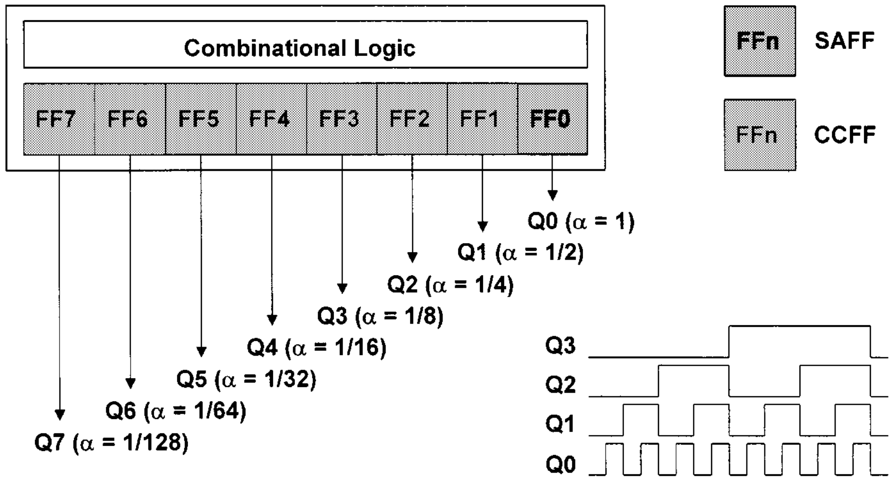

Fig. 7. Block diagram of the counter.

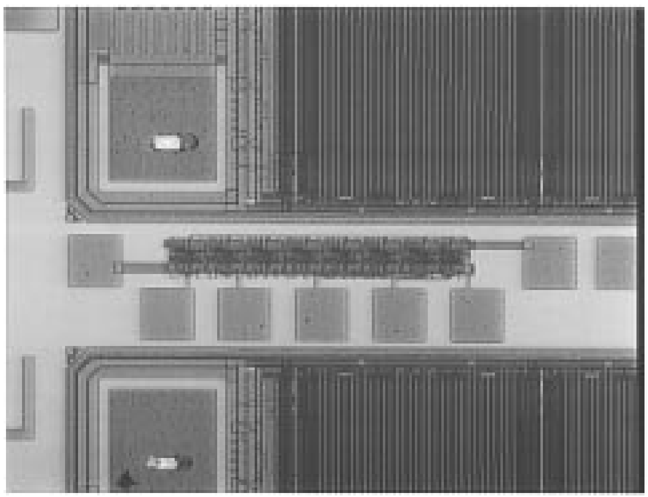

Fig. 8. Microphotograph of the counter.

further reduction on overall power consumption, CCFF can be combined with a number of conventional low-power design methods. For example, various switching probability minimization techniques [14], which have been proposed to reduce the switching portion of the power in logic blocks, can help further reduce overall power consumption. This is achieved by minimizing the flip-flop internal power as well as the logic block switching power by lowering the switching activity of internal nodes including the flip-flop inputs. In the case of using the conventional differential and hybrid flip-flops, only the switching power of the logic block can be reduced because the internal power is relatively independent of the input switching activity. If the switching activities of flip-flops of a digital system are evenly distributed among them and can be estimated at earlier design stages, the flip-flop blending technique [15] can also be helpful. As we have already noted, in terms of power consumption, CCFF is advantageous for inputs with lower switching activity while SAFF is advantageous for inputs with higher switching activity. Thus, for obtaining minimum power consumption, we can select the type of flip-flop with less power for use among many conventional and novel structures. For this, an estimated switching activity profile for each storage element is obtained after completing the design at the logic level. Then, the flip-flop structure that consumes less power can be identified using the activity profile, and thus, can be selected for use by the criterion that SAFF is used for storage elements with higher input switching activities while CCFF for those with lower switching activities. The average switching activity of each storage element in digital systems can be measured using a simulator with suitable input stimulus, or can be estimated at high-level by using power estimation techniques [16], [17]. Some particular circuits like counters have relatively fixed switching activities and thus are readily applicable.

As mentioned earlier, counters are a good example to highlight the low-power advantage of the proposed flip-flops because the switching activities of entire counter bits are known prior and a majority of them are well below 0.5. For the ith bit, the switching activity is $1/2^i$, i.e., only the least significant bit has the maximum switching activity, and the higher the significance of a bit is, the smaller the switching activity becomes. An eight-bit counter, which was designed using the differential CCFF to demonstrate the practical application of the proposed technique, is illustrated in Fig. 7. All the counter bits except for the least significant one were designed using CCFF, while SAFF was used for the latter because the bit has the maximum switching activity. The counter was designed using a 0.35-μm CMOS technology, and the microphotograph is shown in Fig. 8. As shown by Table III, the measured overall power consumption of the counter is well matched with the simulated data. The simulated power consumption of each part of the counter is compared in Fig. 9. The clock and combinational logic power components are comparable for each counter as each flip-flop has the same clock load and combinational logic structure. Although the counter with the proposed flip-flop has additional power component due to extra gates, it has significantly reduced internal power dissipation, and thus, the overall power saving is around 30%.

V. Conclusion

In this paper, novel low-power flip-flops have been introduced. They achieve statistical power reduction by eliminating internal redundant transitions. They have negative setup time and thus provide small data-to-output latency. The overall

TABLE III
POWER CONSUMPTION OF COUNTERS

	Simulated		Measured	
	SAFF	CCFF	SAFF	CCFF
Average power (uW)	951	669	910	640

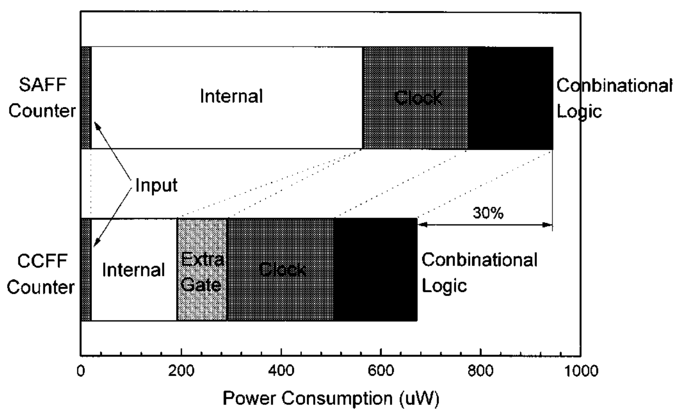

Fig. 9. Comparison of power consumption of counters.

average power consumption of the proposed flip-flops, obtained with a typical input switching activity, is comparable to fully static master–slave designs with the latency comparable to the hybrid and differential designs. The simulation and experimental comparisons indicates that the proposed flip-flops can save a substantial portion of the overall power for low input switching activities. In addition, they can be combined with various switching probability minimization techniques and/or blended with conventional flip-flops for further minimization on overall power consumption.

REFERENCES

[1] S. H. Unger *et al.*, "Clocking schemes for high-speed digital systems," *IEEE Trans. Comput.*, vol. C-35, pp. 880–895, Oct. 1986.

[2] V. Stojanovic *et al.*, "Comparative analysis of master–slave latches and flip–flops for high-performance and low-power systems," *IEEE J. Solid-State Circuits*, vol. 34, pp. 536–548, Apr. 1999.

[3] V. Stojanovic *et al.*, "A unified approach in the analysis of latches and flip-flops for low-power systems," in *Int. Symp. Low-Power Electronics and Design*, Monterey, CA, Aug. 1998, pp. 227–232.

[4] T. Sakurai *et al.*, "Low-power circuit design for multimedia CMOS VLSIs," in *Proc. Synthesis Sys. Integration Mixed Technol. (SASIMI)*, Nov. 1996, pp. 3–10.

[5] H. Partovi *et al.*, "Flow-through latch and edge-triggered flip-flop hybrid elements," in *Int. Solid-State Circuits Conf.e, Dig. of Tech. Papers*, Feb. 1996, pp. 138–139.

[6] F. Klass, "Semi-dynamic and dynamic flip-flops with embedded logic," in *Symp. on VLSI Circuits, Dig. of Tech. Papers*, June 1998, pp. 108–109.

[7] M. Matsui *et al.*, "A 200 MHz 13mm^2 2-D DCT macrocell using sense-amplifying pipeline flip-flop scheme," *IEEE J. Solid-State Circuits*, vol. 29, pp. 1482–1490, Dec. 1994.

[8] B. Nikolic *et al.*, "Sense amplifier-based flip-flop," in *Int. Solid-State Circuits Conf., Dig. of Tech. Papers*, Feb. 1999, pp. 282–283.

[9] B. S. Kong *et al.*, "Conditional-capture flip-flop technique for statistical power reduction," in *Int. Solid-State Circuits Conf., Dig. of Tech. Papers*, Feb. 2000, pp. 290–291.

[10] G. Gerosa *et al.*, "A 2.2 W 80 MHz superscalar RISC microprocessor," *IEEE J. Solid-State Circuits*, vol. 29, pp. 1440–1452, Dec. 1994.

[11] M. Nogawa *et al.*, "A data-transition look-ahead DFF circuit for statistical reduction in power consumption," *IEEE J. Solid-State Circuits*, vol. 33, pp. 702–706, May 1998.

[12] H. Kawaguchi *et al.*, "A reduced clock-swing flip-flop (RCFF) for 63% power reduction," *IEEE J. Solid-State Circuits*, vol. 33, pp. 807–811, May 1998.

[13] J. M. Labaey *et al.*, *Low Power Design Methodologies*. Norwell, MA: Kluwer, 1996, ch. 3.

[14] A. Chandrakasan *et al.*, *Low-Power CMOS Design*. Piscataway, NJ: IEEE Press, 1998.

[15] M. Hamada *et al.*, "Flip-flop selection technique for power-delay trade-off," in *Int. Solid-State Circuits Conf., Dig. of Tech. Papers*, Feb. 1999, pp. 270–271.

[16] A. Ghosh *et al.*, "Estimation of average switching activity in combinational and sequential circuits," in *Proc. 29th Design Automation Conf.*, June 1992, pp. 253–259.

[17] P. Landman *et al.*, "Activity-sensitive architectural power analysis," *IEEE Trans. Computer-Aided Design*, pp. 571–587, June 1996.

Bai-Sun Kong received the B.S. degree in electronics engineering from Yonsei University, Seoul, Korea, in 1990, and the M.S. and Ph.D. degrees in electrical engineering from Korea Advanced Institute of Science and Technology (KAIST), Taejon, Korea, in 1992 and 1996, respectively.

From 1996 to 1999, he was with LG Semicon Corporation Ltd., Seoul, Korea, where he was working on the design of high-bandwidth DRAMs. In 2000, he joined Hankuk Aviation University, Koyang, Korea, where he is currently an Assistant Professor at the School of Electronics, Telecommunication, and Computer Engineering. His research interests include microprocessor and DRAM-related architecture and circuit design, and VLSI circuit and system design for low-power and/or high-speed applications.

Sam-Soo Kim was born in KyungNam-do, Korea, in 1968. He received the B.S. degree in electronic engineering from Kyung-Hee University, Seoul, Korea, in 1993.

In 1993, he joined the DRAM Design Team, LG Semicon. Corporation, Ltd., Seoul, Korea, where he was involved in the design of EDO DRAMs and embedded DRAMs, and he had been engaged in research and development for Rambus DRAM. He is currently working for Hyundai Electronics Corporation, Ltd., Kyungki-do, Korea, as a Research Engineer involved in the development of DDR2 SDRAM. His current interests are high-speed signaling and DRAM architecture.

Young-Hyun Jun received the B.S. degree in electronic engineering from the Han Yang University, Seoul, Korea, in 1984 and the M.S. and Ph.D. degrees in electrical engineering from the Korea Advanced Institute of Science and Technology, Seoul, Korea, in 1986 and 1989, respectively.

He joined Coordinated Science Laboratory of University of Illinois at Urbana-Champaign as a Research Associate in 1989. He joined LG Semicon Corporation, Ltd., Seoul, Korea, in 1991, where he was engaged in DRAM design division as a Director of the Engineering Department. He contributed to the development of 16-Mb and 64-Mb EDO/SDRAM DRAMs and also successfully developed the various generations of Rambus DRAMs including Direct version. He has published many technical papers and holds many patents related to semiconductors. Since 1999, he has worked for SungKyunKwan University, SuWon, Korea, as a Professor of Electrical Engineering. His current interests are in the development of high-speed DRAMs and I/O interfaces, low-power circuits and various analog circuit designs.